

Industrial Internship Report

Project: “Banking Information System”

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BANKING MANAGEMENT SYSTEM

SECURE & EFFICIENT FINANCIAL PLATFORM



DIGITALIZE & OPTIMIZE YOUR OPERATIONS
TRUSTED SOLUTIONS FOR MODERN FINANCE

EXECUTIVE SUMMARY

The **Banking Information System** is a Java-based application developed to provide basic banking operations in a simple, organized, and efficient manner. The main purpose of the project is to computerize common banking activities such as creating customer accounts, depositing money, withdrawing money, checking account balances, and viewing account information. and The project demonstrates how programming concepts can be applied to solve a practical real-world problem. Java concepts such as **classes, objects, methods, conditional statements, loops, input handling, validation, and exception handling** are used in the development of the system.

The project was developed as part of the internship to gain practical experience in software development and problem solving. The system reduces repetitive manual calculations and provides a structured approach for managing banking information. And The current system is a basic learning-oriented prototype. It provides a foundation for future development with **database connectivity, graphical user interfaces, authentication, transaction history, fund transfer, online banking, and enhanced security mechanisms**.

TABLE OF CONTENTS

1. Preface.....	3-5
2. Introduction.....	5-7
○ About UCT	
○ About Upskill Campus	
○ About The IoT Academy	
○ Internship Objectives	
○ References	
○ Glossary	
3. Problem Statement.....	7-10
4. Existing and Proposed Solution.....	10-13
5. Proposed Design/Model.....	13-21
6. Performance Test.....	21-23
7. My Learnings.....	23-25
8. Future Work Scope.....	25-27
9. Conclusion.....	28

1. PREFACE

1.1 Six-Week Work Summary

During the internship period, the Banking Information System was developed through different stages, beginning with understanding the requirements and progressing through programming, design, implementation, testing, debugging, and documentation.

Week 1 – Introduction and Requirement Understanding

The first week focused on understanding the internship requirements, project objectives, and fundamentals of Java programming. The requirements of the Banking Information System were studied and the basic functionalities required for the system were identified.

Week 2 – Java Programming Fundamentals

The second week focused on Java programming concepts required for developing the project. Variables, data types, operators, conditional statements, loops, methods, classes, and objects were studied and practiced.

Week 3 – System Design and Development

During the third week, the basic structure of the Banking Information System was designed. The required banking operations were identified and mapped to suitable program functions.

Week 4 – Feature Implementation

The fourth week focused on implementing major banking operations such as:

- Account creation
- Deposit
- Withdrawal
- Balance enquiry
- Account information display

Week 5 – Testing and Debugging

During the fifth week, the application was tested using different inputs and scenarios. Errors encountered during execution were identified and corrected. Validation was also performed to ensure that banking operations produced the expected results.

Week 6 – Finalization and Documentation

The final week focused on reviewing the source code, organizing the project files, testing the completed application, preparing documentation, uploading the project to GitHub, and preparing the final internship report.

1.2 Importance of Internship

The internship provided an opportunity to gain practical knowledge beyond classroom learning. It helped in understanding the software development process and the importance of planning, implementation, testing, debugging, and documentation.

Working on the Banking Information System helped in applying Java programming concepts to a practical application. The internship improved my problem-solving ability, logical thinking, coding skills, debugging skills, and understanding of software development practices.

1.3 Project / Problem Statement

Traditional banking operations require proper management of customer and account information. Manual handling of banking information can be time-consuming and may result in calculation or record-management errors.

The objective of the Banking Information System is to develop a simple computerized system that can perform basic banking operations efficiently.

The system is designed to allow users to manage account-related information and perform common banking transactions in a structured manner using Java.

1.4 Internship Opportunity

The internship provided practical exposure to software development and programming. It enabled the application of theoretical knowledge gained during academic studies to a practical project.

The Banking Information System served as an opportunity to understand how programming languages, system design, testing, debugging, and version control can be combined to create a software application.

1.5 Planning

The project was planned through the following stages:

1. Understanding the project requirements.
2. Identifying required banking operations.
3. Designing the system structure.
4. Implementing the system using Java.
5. Testing individual operations.
6. Debugging errors.
7. Finalizing the project.
8. Uploading the source code to GitHub.
9. Preparing project documentation.

1.6 Learning and Experience

The project provided practical experience in Java programming and application development.

I learned how to convert a problem statement into a software solution and how to divide a large problem into smaller functions.

The project also provided experience in:

- Java programming
- Object-oriented programming
- Input validation
- Testing
- Debugging
- GitHub
- Technical documentation

1.7 Acknowledgement

I would like to express my sincere gratitude to everyone who supported and guided me during my internship and project development.

I am thankful to my institution, faculty members, internship mentors, coordinators, friends, and family for providing guidance, encouragement, and support throughout the project.

Finally, I am grateful for the opportunity to develop the **Banking Information System**, which helped me improve my technical knowledge, programming skills, logical thinking, and practical understanding of software development.

2. INTRODUCTION

2.1 About UCT

UCT provided the internship framework and learning environment through which students could gain practical exposure to technology and software development.

The internship encouraged students to apply their academic knowledge to practical projects and understand industry-oriented development practices.

2.2 About Upskill Campus

Upskill Campus provides learning and internship opportunities that help students improve their technical knowledge and practical skills.

Through internship activities, students can work on projects, practice programming, develop problem-solving skills, and understand the importance of documentation and project presentation.

2.3 About The IoT Academy

The IoT Academy focuses on providing technology-oriented learning and practical training opportunities. Its programs help students develop technical capabilities and understand how technology can be applied to practical problems.

The internship experience provided an opportunity to develop technical skills and gain practical project experience.

2.4 Internship Objectives

The major objectives of the internship were:

- To gain practical experience in software development.
- To improve Java programming skills.
- To understand object-oriented programming.
- To develop a real-world application.
- To improve problem-solving and logical-thinking skills.
- To understand software testing and debugging.
- To gain experience with GitHub.
- To improve documentation skills.
- To understand the software development process.

2.5 References

The following resources were used for learning and developing the project:

- Java programming learning resources
- Object-oriented programming concepts
- Java development environment documentation
- GitHub documentation
- Internship learning materials
- Programming practice resources

2.6 Glossary

Term	Meaning
Java	Object-oriented programming language used to develop the application
OOP	Object-Oriented Programming
Class	Blueprint used to create objects
Object	Instance of a class

Term	Meaning
Method	Block of code that performs a specific operation
Account	Banking record associated with a customer
Deposit	Adding money to an account
Withdrawal	Removing money from an account
Balance	Amount of money available in an account
GitHub	Platform for hosting and managing source code
IDE	Integrated Development Environment
UI	User Interface
Testing	Process of checking whether the system works correctly

3. PROBLEM STATEMENT

3.1 Problem Definition

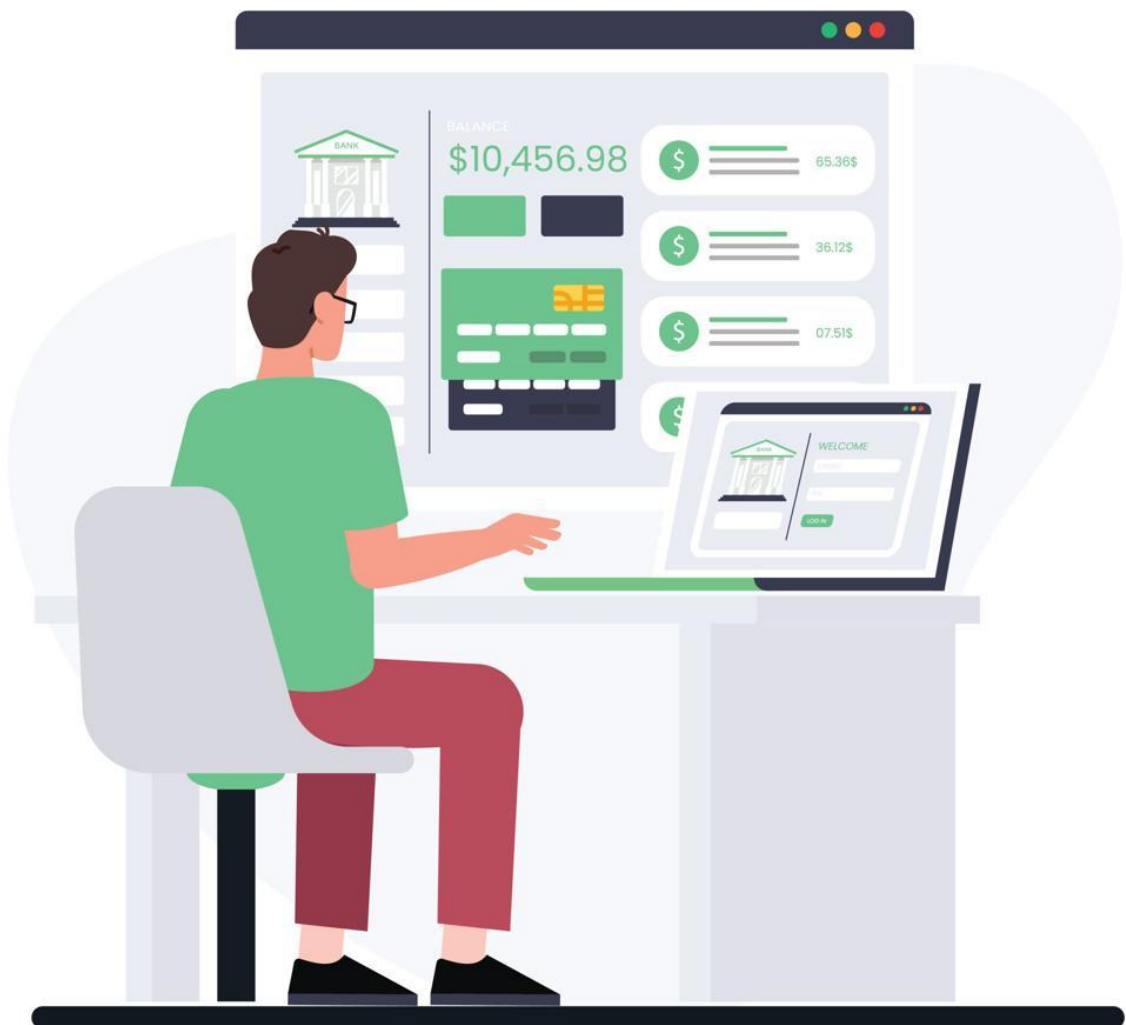
Managing banking information manually can become difficult when customer and transaction information increases.

Manual calculations and record management may lead to:

- Human errors
- Duplicate records
- Time-consuming operations
- Incorrect calculations
- Difficulty maintaining information

Therefore, there is a need for a simple computerized Banking Information System that can perform basic banking operations and manage account information systematically.

The project aims to develop a Java-based application that provides basic banking functionality in a structured and easy-to-use manner.





3.2 Objectives of the Proposed System

The main objectives are:

- To create and manage bank accounts.
- To store customer account information.
- To perform deposit operations.
- To perform withdrawal operations.
- To check account balances.
- To display account information.
- To validate banking operations.

- To reduce manual calculations.
- To provide a simple and organized banking application.

4. EXISTING AND PROPOSED SOLUTION

4.1 Existing Solution

Traditional banking environments may involve manual processes, spreadsheets, or large banking applications.

Manual methods require significant human involvement for record management and calculations.

Large banking systems, on the other hand, require complex infrastructure, databases, authentication systems, network connectivity, and security mechanisms.

4.2 Limitations of Existing Solution

The limitations of a basic manual approach include:

- Time-consuming data management
- Possibility of human errors
- Difficulty maintaining records
- Manual calculation of balances
- Difficult transaction tracking
- Increased effort for repetitive operations
- Limited automation

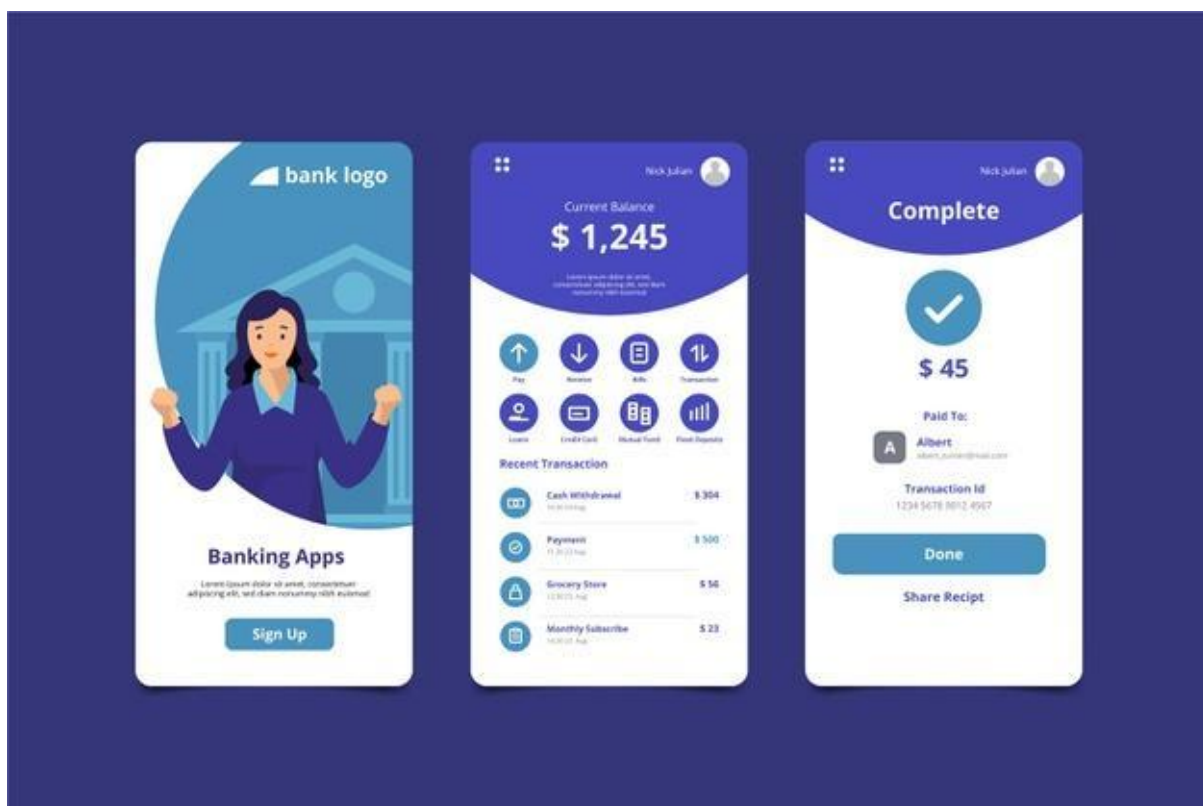
4.3 Proposed Solution

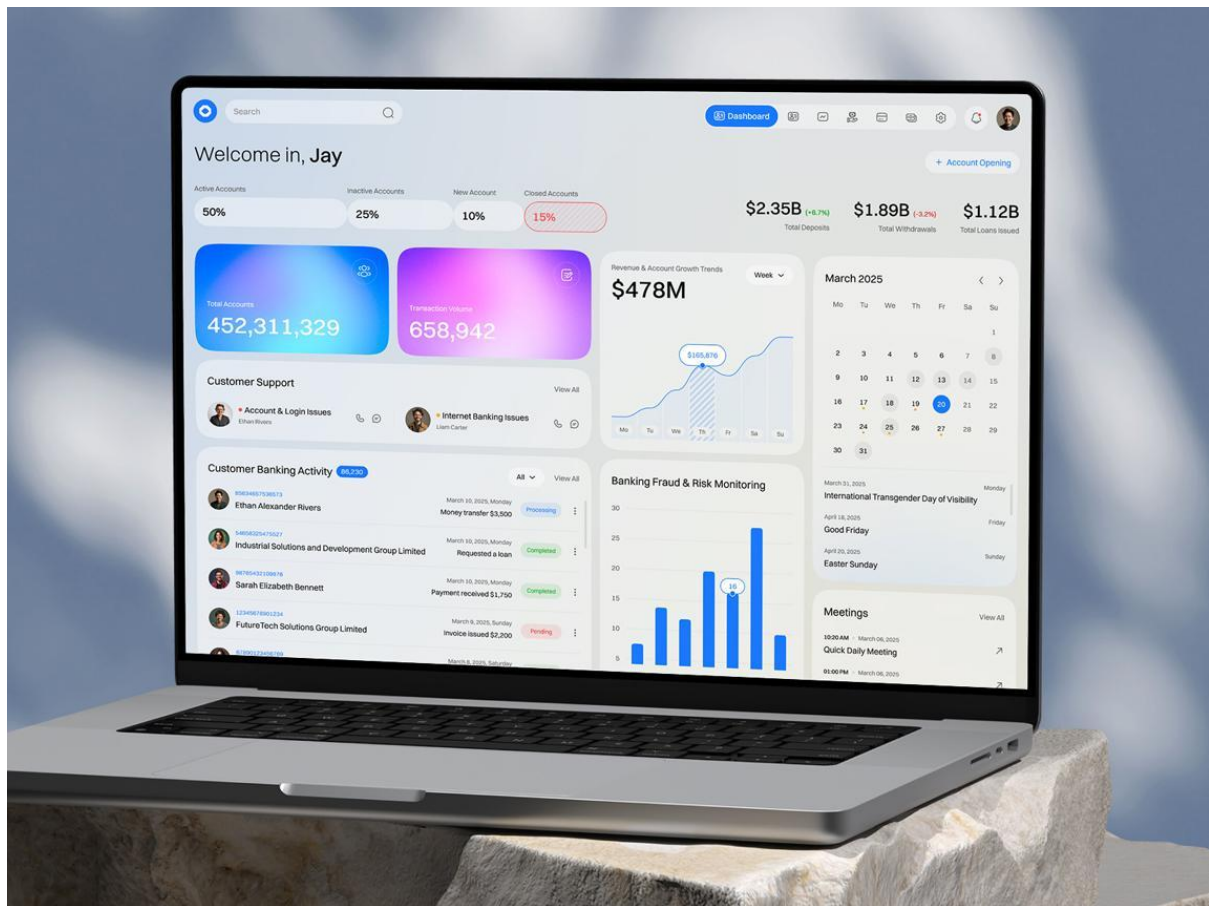
The proposed **Banking Information System** is a Java-based application designed to automate basic banking operations.

The system provides a structured approach for managing customer accounts and performing banking transactions.

The major operations include:

1. Account Creation
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Display Account Information
6. Exit





4.4 Value Addition

The Banking Information System provides the following benefits:

- Reduces manual effort.
- Improves calculation accuracy.
- Provides faster basic operations.
- Organizes account information.
- Provides a simple user interaction process.
- Improves understanding of banking application development.
- Provides a foundation for future expansion.

4.5 GITHUB / CODE LINK

The source code of the Banking Information System is maintained in the following GitHub repository:

GitHub Repository:

<https://github.com/lucky84353646/BankingInformationSystem>

This link can be used by the evaluator to view the project source code and project files.

4.6 REPORT LINK

The internship report is available through the following Google Drive link:

Report Link:

https://drive.google.com/file/d/1iLZODLbTY0qeQzTb7shyd4-oBmT67z_e/view?usp=drive_link

5. PROPOSED DESIGN / MODEL

5.1 System Design

The Banking Information System follows a simple application design.

The user interacts with the banking application and selects the required operation. The application receives the input, validates the information, performs the selected operation, and displays the result.

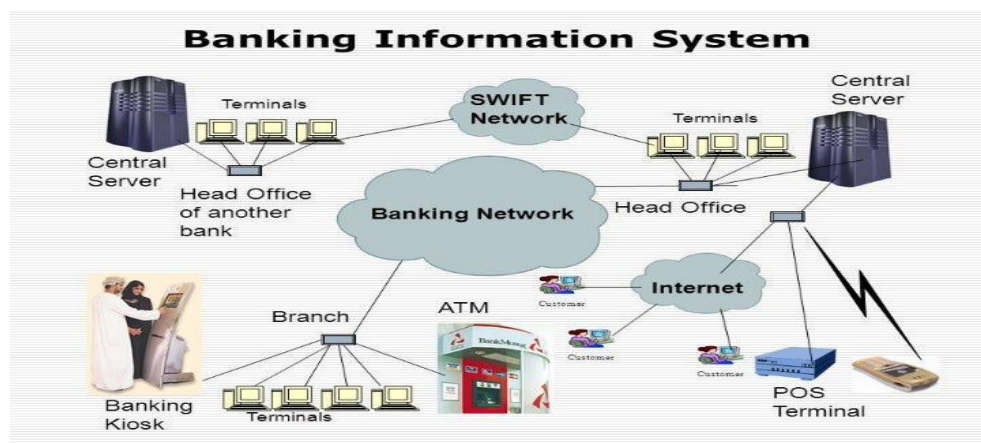
Basic Flow

User → Banking System → Select Operation → Validate Input → Process Request → Update Account Information → Display Result

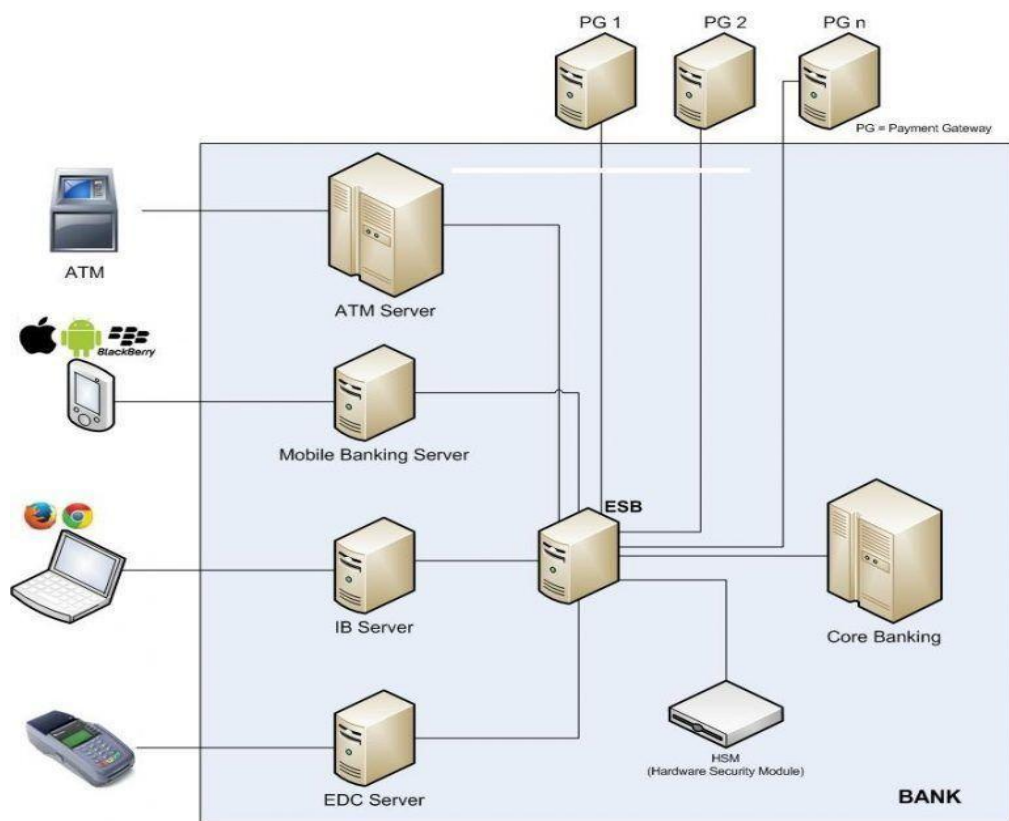
5.2 HIGH-LEVEL DESIGN

The major components of the system are:

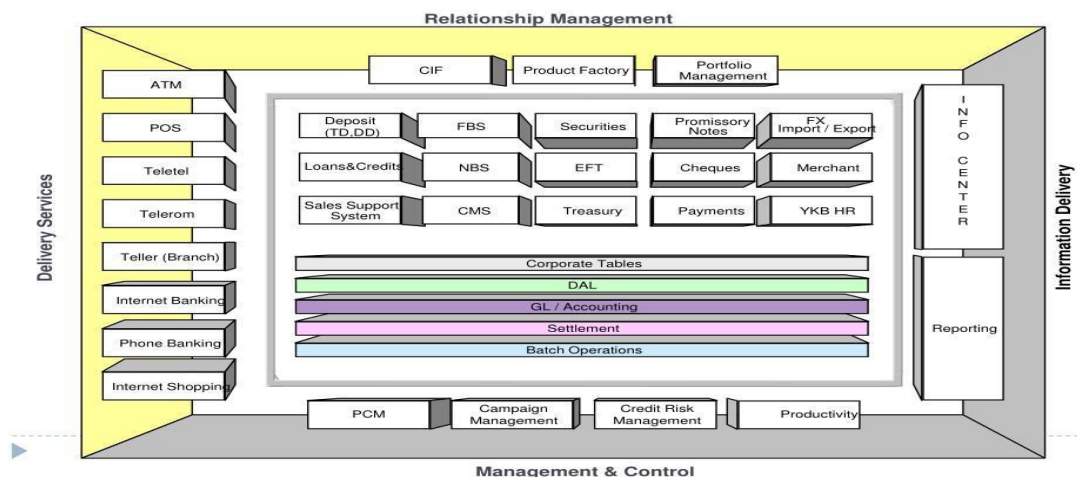
1. **User**
2. **Banking Application**
3. **Account Management**
4. **Transaction Processing**
5. **Information Display**



Architecture



Banking Application Architecture



Architecture Explanation

The user provides input to the Banking Information System. The application processes the request and communicates with the account-management and transaction-processing components.

After processing the operation, the system displays the result to the user.

5.3 LOW-LEVEL DESIGN

The low-level design contains individual programming components used to implement the system.

Typical components include:

- Account class
- Customer/account information
- Deposit method
- Withdrawal method
- Balance checking method
- Account information method
- Main program
- Input validation

These components work together to provide the required banking functionality.

5.4 SYSTEM INTERFACE

The system interface allows the user to interact with the application and select banking operations.

A menu-based interface can provide options such as:

```
=====
BANKING INFORMATION SYSTEM
=====
```

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Display Account Information
6. Exit

Enter your choice:

The user selects an option and provides the required information.

Output

Clear

```
java -cp /tmp/OKQaNwXB57 Bank
Welcome to the Bank of Java1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter an option: 1
Your current balance is $0.0
Welcome to the Bank of Java
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter an option: 2
Enter the amount to deposit: $5000
$5000.0 has been deposited to your account.
```

3 – Deposit funds

If option 3 is selected, your program will ask for an account number and an amount. The confirmation message will state if the deposit succeeded or failed or if the account could not be found

```
Enter account number: 1001(this is an example value)
Enter the amount to deposit: 12.0(this is an example value)
```

The output will be one of the following messages:

Deposit successful, the new balance is: <show new balance>

Or

Deposit failed, the balance is: <show balance>

Or

Account not found

- 1 - Open Account
- 2 - List Accounts
- 3 - Deposit funds
- 4 - Withdraw funds
- 5 - Close an account
- 6 - Exit

Please enter your choice:

5.5 DATA FLOW

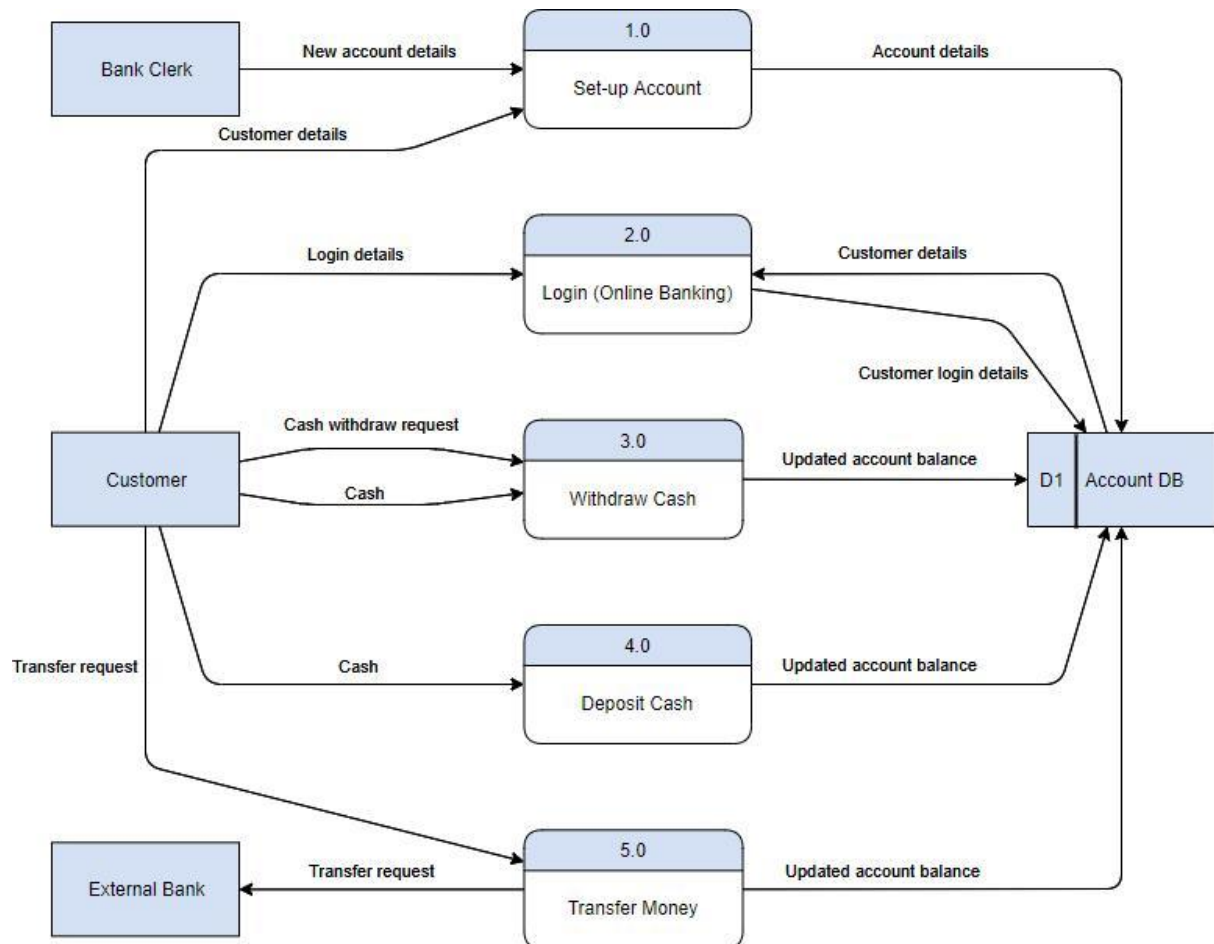
The basic data flow is:

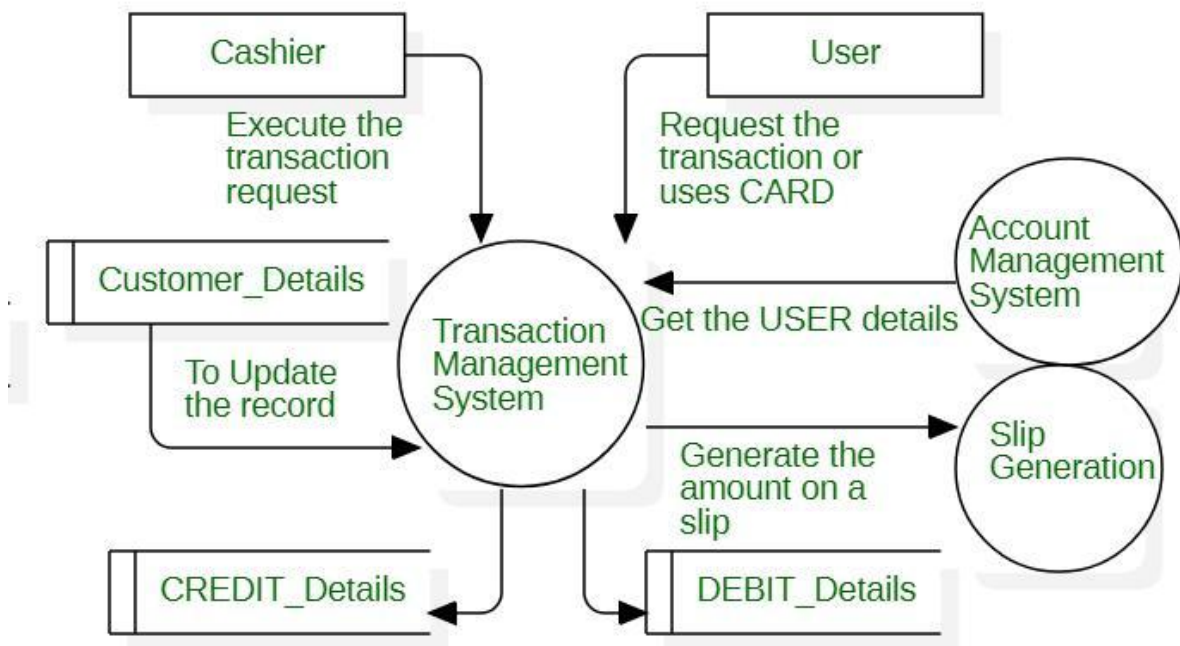
User Input → Validation → Banking Operation → Account Information Update → Output

For example, during a deposit operation:

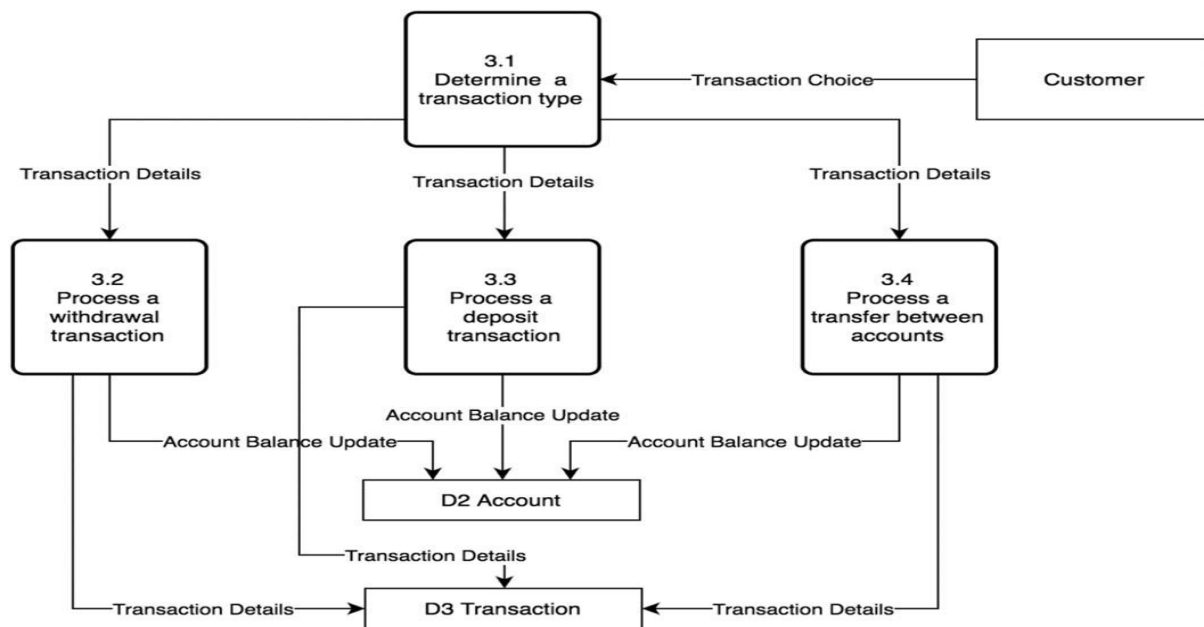
**User → Enter Account Details → Enter Deposit Amount → Validate Amount → Update Balance
→ Display Updated Balance**

Data Flow Image





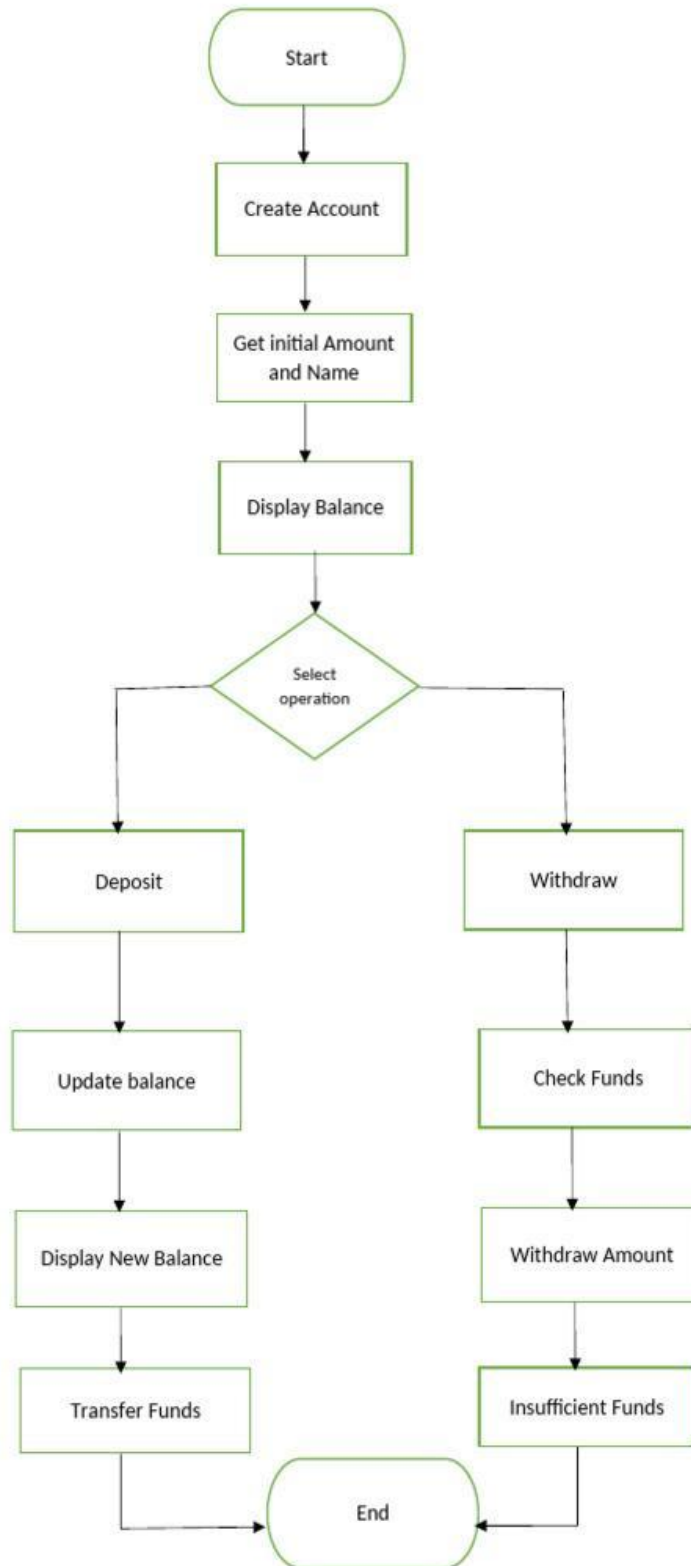
2. Transaction Management System

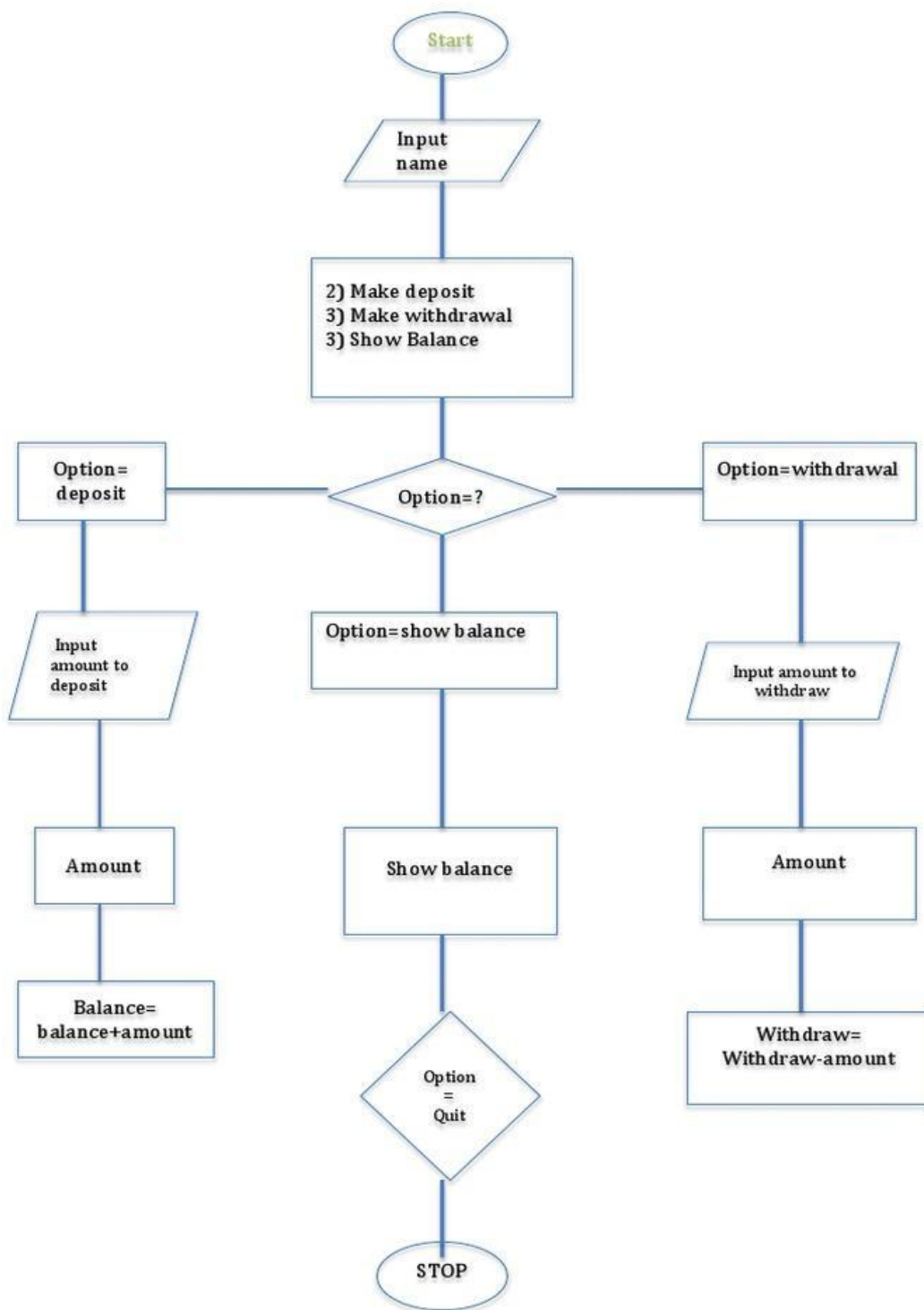


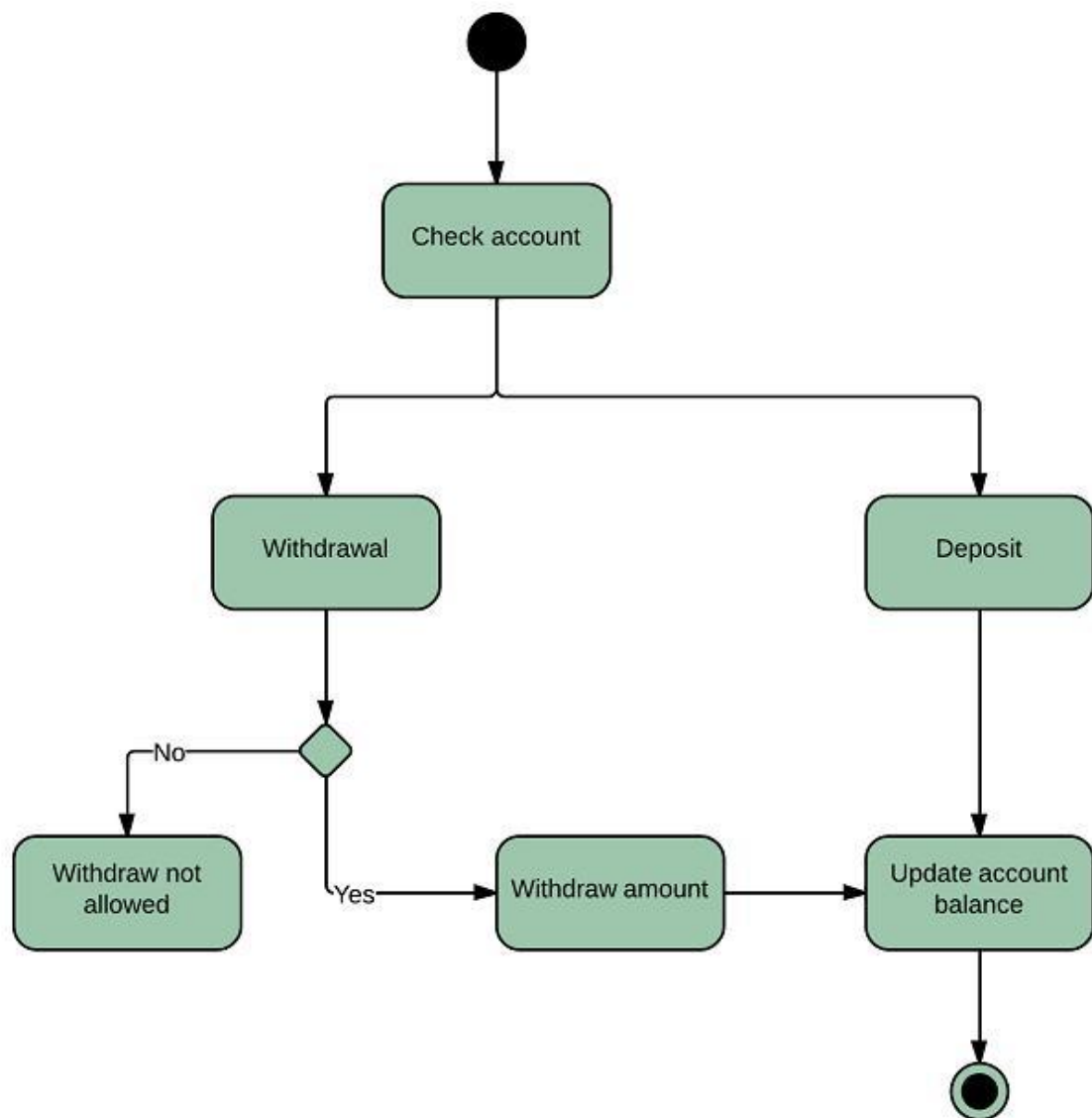
Bank Management System Level 1 DFD for Process 3
(Process a transaction)

5.6 SYSTEM FLOWCHART

The general flow of the application is:







6. PERFORMANCE TEST

6.1 Constraints

The project has certain limitations because it is designed as a basic banking application.

The major constraints include:

- Limited to basic banking operations.
- No real banking network integration.
- Limited persistent data storage if no database is used.
- No real-time online transaction processing.
- Basic security compared with commercial banking systems.
- Limited user interface functionality.

6.2 TEST PLAN

The system was tested using different inputs and scenarios to verify that the banking operations work correctly.

The major test areas include:

- Account creation
- Deposit operation
- Withdrawal operation
- Balance enquiry
- Account information display
- Invalid input handling
- Insufficient balance handling
- Application exit

6.3 TEST CASES

Test Case	Input / Operation	Expected Result
TC01	Create account with valid details	Account should be created
TC02	Deposit valid amount	Balance should increase
TC03	Withdraw valid amount	Balance should decrease
TC04	Withdraw amount greater than balance	Transaction should be rejected
TC05	Check account balance	Current balance should be displayed
TC06	Display account information	Account information should be displayed
TC07	Enter invalid amount	Appropriate validation should occur
TC08	Exit application	Application should terminate

6.4 TEST PROCEDURE

The application was executed and each available operation was tested individually.

Valid and invalid inputs were provided to verify the behavior of the system.

The actual output was compared with the expected output. Errors identified during testing were corrected through debugging and the application was tested again.

6.5 PERFORMANCE OUTCOME

The Banking Information System is designed to perform basic operations quickly because the core processing is performed locally within the Java application.

The testing process helps verify:

- Correct account operations
- Correct deposit calculations
- Correct withdrawal calculations
- Balance checking
- Account information display
- Invalid input handling
- Insufficient balance handling

A production banking application would require additional performance testing, database testing, security testing, concurrency testing, and transaction-reliability testing.

7. MY LEARNINGS

7.1 Technical Learnings

During the development of the Banking Information System, I gained knowledge of:

- Java programming
- Object-oriented programming
- Classes and objects
- Methods
- Conditional statements
- Loops
- Variables and data types
- Input handling
- Validation
- Error handling
- Program testing
- Debugging
- GitHub
- Source-code management

7.2 Practical Learnings

The project helped me understand how a software project is developed from an initial problem statement to a working application.

I learned the importance of planning the system before writing code.

I also learned how individual programming components can be combined to create a complete application.

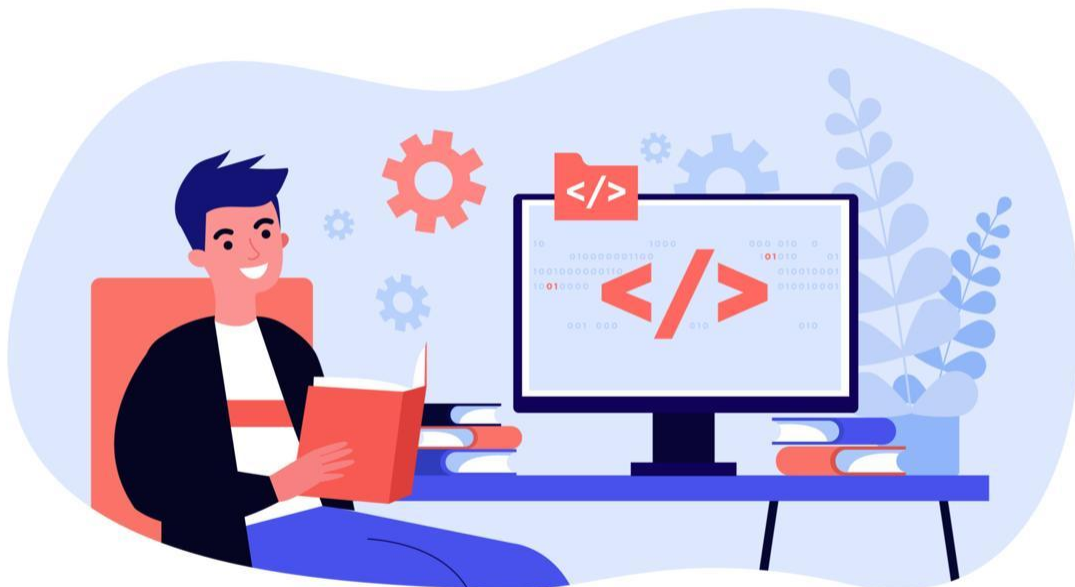
Testing and debugging helped me understand how to identify errors and improve the reliability of an application.

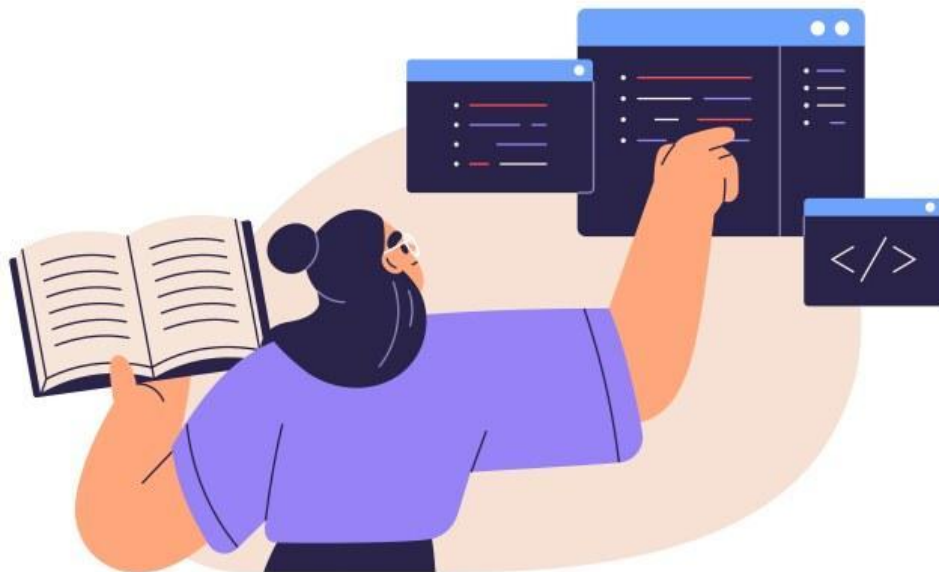
7.3 Career Benefits

The internship and project experience will help me in my future career by improving my programming and problem-solving abilities.

The Banking Information System provides practical experience that can be useful for future software-development projects.

The knowledge gained from Java, GitHub, testing, debugging, and project documentation can also help in preparing for technical interviews and software development roles.





8. FUTURE WORK SCOPE

The Banking Information System can be enhanced in several ways in the future.

8.1 Database Integration

A database such as MySQL can be integrated to permanently store customer, account, and transaction information.

8.2 Graphical User Interface

A graphical interface can be developed using technologies such as JavaFX or Swing to make the application more user-friendly.

8.3 Authentication

A secure login system can be added using usernames, passwords, PINs, or multi-factor authentication.

8.4 Transaction History

The system can maintain complete transaction history for deposits, withdrawals, transfers, and other operations.

8.5 Fund Transfer

A fund-transfer feature can be introduced to allow users to transfer money between accounts.

8.6 Online Banking

The application can be extended into a web or mobile banking system that allows users to access their accounts remotely.

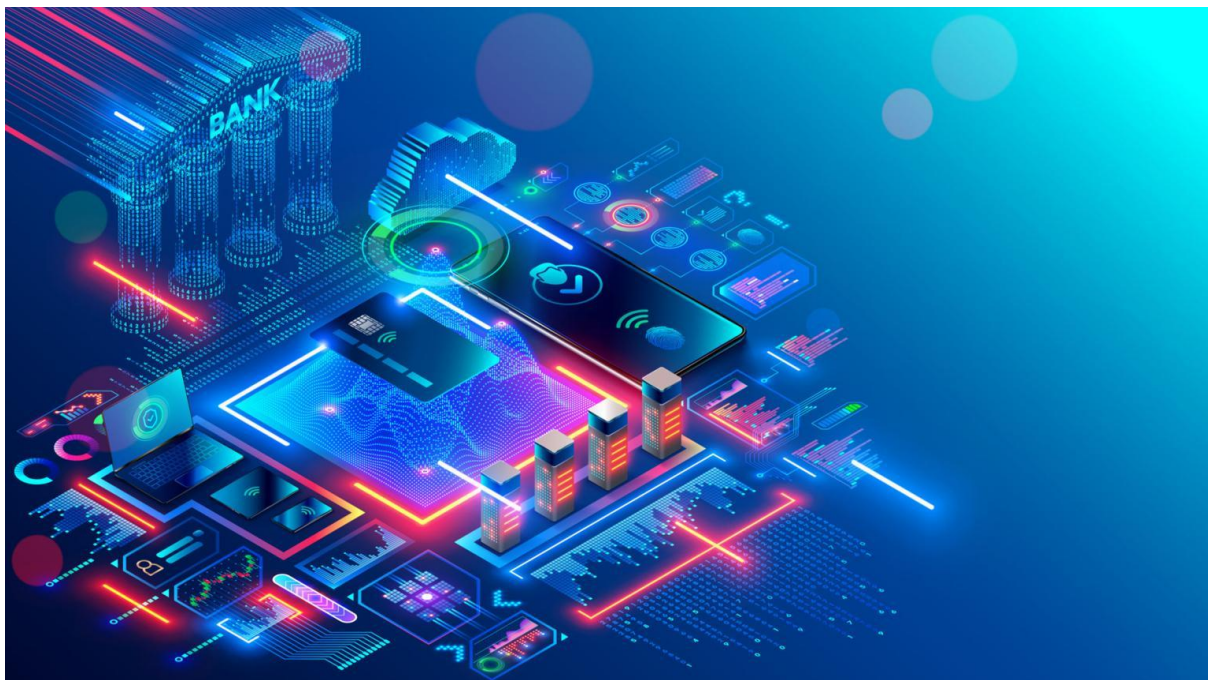
8.7 Enhanced Security

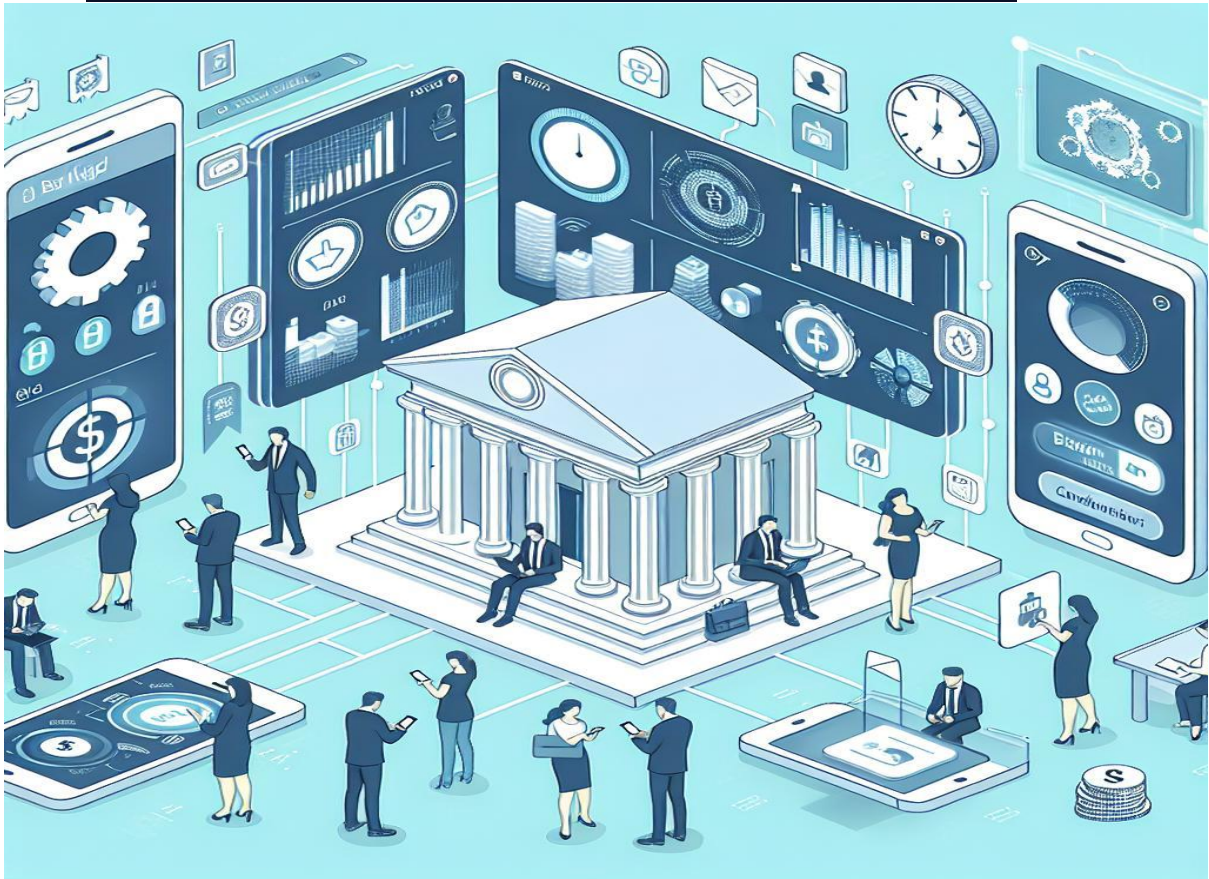
Encryption, secure authentication, authorization, and other security mechanisms can be implemented to protect sensitive banking information.

8.8 Advanced Reporting

The system can generate account statements, transaction reports, and other useful banking reports.

Future Scope Image





9. CONCLUSION

The **Banking Information System** project provided practical experience in developing a software application using Java. The project focused on implementing basic banking operations in a structured and efficient manner.

Through this project, I gained practical knowledge of programming, object-oriented concepts, system design, testing, debugging, and GitHub-based project management.

Although the current system focuses on basic banking operations, it provides a strong foundation for future development. Database integration, graphical interfaces, authentication, transaction history, fund transfers, and advanced security can be added to transform the application into a more complete banking management system.

Overall, the project successfully helped bridge the gap between theoretical programming knowledge and practical software development experience.

PROJECT SUBMISSION LINKS

GitHub Project

<https://github.com/lucky84353646/BankingInformationSystem>

Internship Report

[https://drive.google.com/file/d/1iLZODLbTY0qeQzTb7shyd4-oBmT67z_e/view?usp=drive link](https://drive.google.com/file/d/1iLZODLbTY0qeQzTb7shyd4-oBmT67z_e/view?usp=drive_link)